

Mark schemes

Q1.

- (a) Resolve vertically:

$$T \cos \theta = mg$$

M1 for $T \cos \theta$ or $T \sin \theta$ and mg

M1

$$34 \cos \theta = 2 \times 9.8$$

A1

$$\cos \theta = \frac{19.6}{34}$$

$$\theta = 54.8^\circ$$

A1

3

- (b) Resolve horizontally for particle:

$$\frac{mv^2}{r} = T \sin \theta$$

M1 for $T \cos \theta$ or $T \sin \theta$

M1

$$v^2 = \frac{34 \sin 54.8 \times 0.8}{2}$$

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A1 ft
from
(a)

$$v^2 = 11.113$$

Speed is 3.33 m s^{-1}

Accept 3.34

A1

3

- (c) Time taken is $2\pi r / v$

Or find ω and use $\frac{2\pi}{\omega}$

M1

$$= 1.51 \text{ sec}$$

Q2.

- (a) Resolving vertically

$$T \cos 30 + 20 \cos 50 = 4g$$

M1: Three terms, which must include $4g$, $T \cos \theta$ or $T \sin \theta$ and $20 \cos \theta$ or $20 \sin \theta$, where $\theta = 30, 40, 50$ or 60 .

A1: Correct terms

A1: Correct equation

M1A1

A1

$$T \cos 30 = 26.344$$

$$T = 30.4 \text{ N}$$

A1: Correct final answer.

Accept 30.4 or AWRT 30.42.

Accept 30.4 or 30.5 or AWRT 30.45 from $g = 9.81$.

A1

4

- (b) Horizontally: $\frac{mv^2}{r} = 20 \cos 40 + T \cos 60$

M1: Three terms, which must include $\frac{mv^2}{r}$ or $\frac{4 \times 5^2}{r}$, $T \cos \theta$ or $T \sin \theta$ and $20 \cos \theta$ or $20 \sin \theta$, where $\theta = 30, 40, 50$ or 60 .

A1F: Correct equation. May include T , m and v .

M1

A1F

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$$\frac{4 \times 5^2}{r} = 30.53$$

dM1: Substitution of values for T , m and v .

Equation of form $\frac{4 \times 5^2}{r} = \text{number}$

dM1

$$r = 3.27537$$

$$= 3.28$$

A1: Correct answer. Accept 3.27 or 3.28 or AWRT 3.28.

Accept 3.27 or AWRT 3.27 from $g = 9.81$.

Note: Do not accept $\frac{mv^2}{r} = 30.4$ or similar.

A1

4

Q3.

(a) $r = 1.2 \sin \theta$

$1.2 \cos \theta$ 0 marks

B1

1

(b) Resolve horiz: $T \sin \theta = m\omega^2 r$

$T \cos \theta = m\omega^2 r$ etc M1 (+ second M1)

M1A1

$$T \sin \theta = 4 \times 5^2 \times 1.2 \sin \theta$$

$$T = 120$$

A1

Resolve vert: $T \cos \theta = 4g$

$$\theta = \frac{30 \sin \theta}{g}$$

M1 for tan

M1A1

$$\cos \theta = 0.32666$$

$$\theta = 70.9^\circ \text{ or } 1.24^\circ$$

A1

6

[7]

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Q4.

(a) 40 revolutions per minute

= 80π radians per minute

$$\text{or } \frac{2}{3} \text{ rev per second}$$

B1

$$= \frac{4\pi}{3} \text{ radians per second}$$

AG

B1

2

(b) Resolve vertically:

$$T \cos 30 = 6g$$

M1 1 term each side, 1 correct

M1A1

$$T = 67.9 \text{ N}$$

AG

A1

3

- (c) Resolve horizontally:

$$T \sin 30 = m\omega^2 r$$

M1 1 term each side, 1 correct

M1

$$A1 \ T \sin 30$$

A1

$$67.9 \sin 30 = 6 \times r \times \left(\frac{4\pi}{3}\right)^2$$

A1 RHS

A1

$$r = 0.322 \text{ m}$$

Condone 0.323 (using π as 3.14)

A1

4

[9]

EXAM PAPERS PRACTICE

Q5.

- (a) Resolving vertically:

$$T \cos 60 + T \cos 40 = mg$$

M1A1

$$1.266 T = 6g$$

M1

$$T = 46.4 \text{ N}$$

AG no marks if g deleted

A1

4

- (b) Radius of circle is $0.6 \tan 60$

$$r = 1.039 \text{ or } 1.04$$

B1

Horizontally:

$$\frac{mv^2}{r} = T \cos 50 + T \cos 30$$

Accept sin instead of cos for M1

M1

$$\frac{6v^2}{1.039} = 46.4 \cos 50 + 46.4 \cos 30$$

or 70.01

A1

$$v^2 = 12.123$$

Speed is 3.48 m s^{-1}

A1

4

[8]

Q6.

(a) Acceleration is $\frac{v^2}{r}$

$$= \frac{2^2}{0.2}$$

M1

$$= 20 \text{ m s}^{-2}$$

A1

2

(b) $\theta = 30^\circ$

B1

Resolve vertically:

$$T_1 \cos \theta = mg$$

M1

$$T_1 \cos \theta = 4g$$

A1

$$T_1 = 45.3 \text{ N}$$

AG

A1

4

- (c) Resolve horizontally:

$$T_1 \sin \theta + T_2 = \frac{mv^2}{r}$$

M1 for 3 terms, 2 correct

M1A1

$$45.3 \sin \theta + T_2 = 4 \times 20$$

$$T_2 = 57.4 \text{ N}$$

Condone 57.3 N

A1

3

[9]

Q7.

(a) $\frac{40 \times 2\pi}{60} = \frac{4\pi}{3} \text{ (rad/sec)}$

M1A1

2

(b) $\alpha = \omega^2 r = \left(\frac{4\pi}{3}\right)^2 \times 0.2$

M1

$$= \frac{16\pi^2}{45}$$

Accept $0.356\pi^2$ (3sf)

A1

2

- (c) (i)



B1

1

- (ii) **Vertically**
 No acceleration, forces balance
 $mg = T \cos \theta$

B1

1

- (iii) **Horizontally**

$$T \sin \theta = m \times \frac{16\pi^2}{45}$$

M1

ft acceleration

A1F

$$T \cos \theta = mg$$

$$SC \tan \theta = \frac{\omega^2 r}{g}$$

using correctly

1st 3 marks for quoting and

m1

$$\tan \theta = \frac{16\pi^2}{45g}$$

or

$$\tan \theta = 0.358 \text{ (08)}$$

ft provided M1

A1F

$$\theta = 20^\circ$$

earned in (b)

A1F

5

[11]

Q8.

- (a) Q is in equilibrium

Q at rest, or not moving

E1

$$T = 5g = 49 \text{ N}$$

AG

B1

2

- (b) Resolving vertically for P:

$$T \cos \theta = 3g$$

M1A1

$$\cos \theta = \frac{3}{5}$$

$$\theta = \cos^{-1} \frac{3}{5} = 53.1^\circ$$

Do not condone 53°

A1

3

(c) $\therefore \sin \theta = \frac{4}{5}$

B1

Resolving horizontally for P:

$$\frac{mv^2}{r} = T \sin \theta$$

M1 2 terms: 1 term correct, other term includes sin or cos

M1A1

$$\frac{3v^2}{r} = \frac{4}{5} \times 5g$$

$$\frac{3 \times 4^2}{r} = 4g$$

$$r = \frac{48}{4g}$$

$$= 1.22$$

SC3 1.23

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A1

4

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[9]

Q9.

(a) $T \cos 30^\circ = 2 \times 9.8$

Resolving vertically with two terms

M1

Correct equation

A1

$$T = \frac{2 \times 9.8}{\cos 30^\circ}$$

$$T = 22.6 \text{ N} \quad \text{AG}$$

Correct T from correct working

A1

$$(b) \quad T \cos 60^\circ = 2 \times \frac{v^2}{0.6}$$

M1

*Resolving horizontally.
Correct equation*

A1

Solving for v

dM1

$$v = 1.84 \text{ ms}^{-1}$$

Correct v

A1

4

[7]

Q10.

$$(a) \quad (i) \quad T \cos 60^\circ = 0.4 g$$

M1A1

$$T = 7.84 \text{ newtons}$$

AG

A1

3

$$(b) \quad T \cos 30^\circ = 0.4 \omega^2 r$$

M1A1

$$r = \tan 60^\circ$$

B1

$$7.84 \times \cos 30^\circ = 0.4 \omega^2 \times \tan 60^\circ$$

Subs

m1

$$\omega = 3.13$$

AG

A1

5

$$(c) \quad T = \frac{2\pi}{\omega} = 2.007$$

$\approx 2 \text{ sec}$

B1

1

[9]

Q11.

(a) Period = $\frac{2\pi}{\omega}$

M1

$$\therefore 1.5 = \frac{2\pi}{\omega}$$

$$\omega = \frac{2\pi}{1.5} = \frac{4\pi}{3} = 1.3\pi$$

Any – must leave π

A1

2

(b) (i) Acceleration = $r\omega^2$

$$= 0.6 \left(\frac{4\pi}{3} \right)^3$$

Attempt to use formula

M1

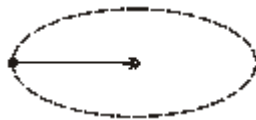
$$10.5 \text{ or } \frac{16\pi^2}{15}$$

Follow through their ω

A1ft

2

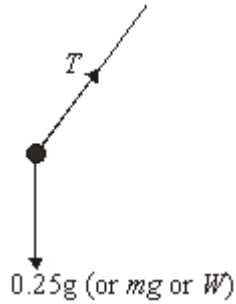
(ii)



B1

1

(c) (i)



(b)(ii) & (c)(i) could be on a single diagram

B1

1

(ii) Vertically (let $\hat{AO} = \alpha$) $T \sin \alpha = mg$ (1)

Horizontally

Values may or may not be substituted in each equation throughout

$$T \cos \alpha = m r \omega^2 \quad (2)$$

ft $r \omega^2$ from b(i)

M1A1

M1A1ft

$$(1) \div (2) \quad \tan \alpha = \frac{g}{r \omega^2}$$

Dividing to get $\tan \alpha$ or square and add to get T first

M1

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$$\alpha = \tan^{-1} \frac{9.8}{0.6 \left(\frac{4\pi}{3} \right)^2} \quad \text{or} \quad \tan^{-1} (0.9308)$$

$$\alpha = 43^\circ$$

Rounds to 43°

A1

6

(d) No air resistance
Modelled as a particle

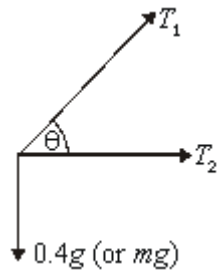
E1

1

[13]

Q12.

(a)



Three forces evident;
 T_1 , T_2 clear – on diagram or in calculations

B1

1

(b) Vertically $T_1 \sin \theta = 0.4g$

Resolve vertically – component evident
 $T \sin \theta = mg$ seen

$$T_1 \left(\frac{4}{5} \right) = 0.4g$$

$$T_1 = 0.5g = 4.9\text{N}$$

AG obtained

M1 A1

M1

A1

3

(c) $T_2 = 0.1k$

B1

1

(d) $a = r\omega^2 = 0.3 \times 5^2$

Calculation of a seen

B1

$$\text{Force} = T_2 + T_1 \cos \theta$$

(Both terms of horizontal resultant force attempted)

M1

$$= 0.1k + 4.9 \times \frac{3}{5}$$

Previous expression / result substituted

A1

$$F = ma$$



m1

$$k = 0.6$$

AG

A1

5

(e)
$$\text{EPE} = \frac{kx^2}{2} = \frac{0.6(0.1)^2}{2}$$

$$\frac{kx^2}{2} \text{ seen and attempt to use}$$

M1

$$= 0.003 \text{ J}$$

A1

2

[12]

Q13.

(a) Horizontally

$$T \sin \theta = m r \omega^2$$

Newton's Law horizontally

M1A1

$$r = l \sin \theta$$

$$\Rightarrow T \sin \theta = m l \sin \theta \omega^2$$

Use of $r = l \sin \theta$ in equation

B1

$$\Rightarrow T = m l \omega^2$$

Cancelling to obtain answer

A1

4

(b) Vertically

$$T \cos \theta = mg$$

B1

$$\Rightarrow (m l \omega^2) \cos \theta = mg$$

Use of part (a)

m1

$$\Rightarrow \cos \theta = \frac{g}{l\omega^2}$$

A1

3

(c) (i) $T \leq 16$

$$\Rightarrow (0.1)(0.4)\omega^2 \leq 16$$

Use of part (a)

M1

$$\omega \leq 20$$

Withhold A1 if inequalities not evident

(use of $\omega_{\max}$ or clearly explained conclusion accepted)

A1

2

(ii) Substitute in (b)

$$\theta = \cos^{-1} \left(\frac{9.8}{0.4(20^2)} \right)$$

Substitute in formula and use of $\cos^{-1}$

M1

$$= 86^\circ$$

CAO

A1

2

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[11]

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Q14.

(a) $a = \frac{4^2}{0.1} = 160 \text{ ms}^{-2}$

Calculating the acceleration

M1

Correct acceleration

A1

2

(b) $\cos \theta = \frac{\sqrt{30}}{20} = \frac{\sqrt{3}}{2}$

Identifying trig value

B1

$$T_1 \cos \theta = 2 \times 9.8$$

Resolving horizontally

M1

Correct equation

A1

$$T_1 = \frac{2}{\sqrt{3}} \times 2 \times 9.8 = 22.6 \text{ N}$$

Correct force

A1

4

(c) $\sin \theta = \frac{1}{2}$

Identifying trig value

B1

$$T_1 \sin \theta + T_2 = 2 \times 160$$

Resolving horizontally

M1

Correct equation

A1

$$T_2 = 320 - \frac{39.2}{\sqrt{3}} \times \frac{1}{2} = 309 \text{ N}$$

Substituting and solving

m1

Correct force

A1

5

[11]

Q15.

(a) (i) $T \cos 30^\circ = 3 \times 9.8$

Resolving vertically

M1

$$T = 33.9 \text{ N}$$

Correct equation

Correct tension (awrt 33.9)

A1 A1

3

(ii) $r = 0.6 \times \sin 30^\circ = 0.3$

Correct radius

B1

$$33.9 \cos 60^\circ = 3 \times 0.3 \omega^2$$

Resolving horizontally

Solving for ω

m1 m1

$$\omega = 4.34 \text{ rad s}^{-1}$$

Correct ω from correct working

m1 A1

4

(b) $T = \frac{3 \times 9.8}{\cos \theta}$

Resolving vertically to get a correct expression

Resolving horizontally

B1 M1

$$\frac{3 \times 9.8}{\cos \theta} \sin \theta = 3 \times 0.6 \sin \theta \times 8.68^2$$

Use of correct ω^2

correct equation

A1 A1

$$\cos \theta = \frac{9.8}{0.6 \times 8.68^2}$$

Solving for θ using $\cos \theta$

m1

$$\theta = 77^\circ$$

cao

A1

6

[13]

Examiner reports

Q1.

In part (a) many students resolved vertically to find the angle but in the required equation $T \cos \theta = mg$ a number of students used $\sin \theta$ rather than $\cos \theta$ whilst others used $mg = \frac{T}{\cos \theta}$. Similar errors were seen in part (b). In part (c) many students found ω and then stopped. The value of ω was not credited since it was just as simple to find the time from knowing only v as it was from knowing ω .

Q2.

Many candidates answered this question well. Part (a) was usually correct but a few candidates only considered the component of one of the tensions to equate to mg . Most candidates answered part (b) well.

Q3.

Part (a) was usually answered well. However a significant proportion of candidates did not resolve in part (b) and hence were unable to find θ .

Q4.

In part (a), the conversion of 40 revolutions per minute into $\frac{4\pi}{3}$ radians per second was usually shown correctly. In part (c), candidates often forgot the m in the equation $T \sin 30 = \frac{mv^2}{r}$. A common error was in the use of $T \sin 30 = \frac{mv^2}{r}$, where candidates forgot that they were given the value of ω and not the value of r . Candidates were often unable to find correctly the numerical value of r when they had correctly obtained $r = \frac{67.9 \sin 30}{6 \times \left(\frac{4\pi}{3}\right)^2}$. Some gave the answer as 0.32 and it was surprisingly common for candidates to use $\pi = 3.14$ to find an answer of 0.323. Although this was accepted, it would have been far simpler for candidates to use the π button on their calculators.

Q5.

This question caused difficulty to some candidates as there were two strings to consider. Resolving vertically did not always give them the equation $T \cos 60 + T \cos 40 = mg$. Others used the result 46.4, given in the question to help candidates, and hence appreciated that they needed to add the two components of the tension. Unfortunately, many candidates then returned to their knowledge of more typical questions and only considered the component of one string when resolving horizontally.

Q6.

Parts (a) and (b) were usually done well. However, in part (c), many candidates did not

show a full force diagram, only giving $T_{PB} = \frac{mv^2}{r}$, omitting the term $T_{PA}\sin\theta$.

Q7.

Part (a) was popular and done well. Part (b) often began well but many candidates were unable to find a correct and suitably simplified expression. A significant number did not know a correct formula for the acceleration, or confused v and ω . Parts (c)(i) and (c)(ii) were mostly done well, and in part (c)(iii) those with correct radial equations usually completed the question successfully.

Q8.

In part (a), candidates were asked to **explain** why $T = 49\text{N}$. Many candidates simply showed that $5g = 49$. The fact that particle Q was in equilibrium, or at least was not moving, needed to be stated. Most candidates answered parts (b) and (c) correctly, but there were some instances where three forces appeared in the resolving equations in either or both of these parts.

Q9.

There were some good responses to this question. In part (a), a small number of candidates did not show sufficient working to justify the marks available, even though they were able to obtain the correct answer. It is important in questions of this type to demonstrate that any calculations result from resolving or considering a triangle of forces. Some of the poor responses looked as if the candidates had experimented with the numbers until they had obtained 22.6. The responses to part (b) were generally very good or completely unsuccessful. Interestingly, some candidates did part (a) but made no progress with part (b), while some others were unable to do part (a), but used the printed answer to enable them to do part (b).

Q11.

This question provided a better response than was expected. The structure seemed to help candidates to score well. A number of candidates did not seem to be aware of how to find the angular speed given angular displacement and time – part (a) therefore proved discriminating. The correct formula was often seen in part (b), although some substituted incorrectly. Diagrams were excellent and well labelled. Some candidates were well drilled in finding the angle, although some found the angle with the vertical because they had used the angle at the top. Follow through marks meant a critical error early on often only lost

Q12.

This proved to be a demanding question although a significant proportion of candidates solved it well. The force diagram in part (a) was rarely correct with equal tensions shown or, even worse, the tension in OP missing and sometimes a normal reaction appearing at P . Those candidates who had not clearly marked different tensions were able to recover the mark if their working out indicated the use of two different tensions.

In parts (b) and (d) many candidates struggled to set up the equations required. The answer in part (b) was arrived at through incorrect methods to match the answer given. In part (d) mrw^2 was often seen, but a number of candidates thought that this was directed along QP or outwards from P .

Parts (c) and (e) were more successful and it was pleasing to see the correct formulae in use.

Q13.

Candidates who were familiar with the relevant book work performed well in this question. It was clear, though, that many candidates had not studied such work in sufficient depth. Many answers were faked to obtain the printed answer in part (b).

It was disappointing that some candidates could not substitute correct values in to the equation in part (b) to obtain the 2 marks in part (c) (ii).

Q14.

The candidates found part (a) straightforward and there were relatively few errors here. A few candidates calculated the resultant force rather than the acceleration. In parts (b) and (c), a few had trouble finding the angles to use when resolving, but most of those who made a reasonable attempt did manage to use the correct angles. There were some errors in forming the equations of motion, often due to either including additional terms or making errors when resolving.

Q15.

Part (a) was generally done well. The most common error was to resolve the weight instead of the tension, to give an equation such as $3g \cos 30 = T$. Many candidates were able to obtain the given value of ω , but in some cases the working was not detailed enough. Some candidates simply quoted the result $T = ml\omega^2$ and did not show how to obtain this result by resolving the tension. The candidates found part (b) more demanding and only the better candidates were able to complete the solution. The most common error arose from failing to substitute for the radius correctly, so that these candidates obtained a value for $\tan \theta$ rather than $\cos \theta$.